

Secondary Examination, 2016

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 80

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) Candidate must write first his/ her Roll No. on the question paper compulsorily.
- 2) All the questions are compulsory.
- 3) Write the answer to each question in the given answer- book only.
- 4) For questions having more than one part the answers to those parts are to be written together in continuity.
- 5) If there is any error/difference/contradiction in Hindi and English versions of the question paper of Hindi version be treated valid.
- 6)

Section	Q.Nos.	Marks per question
A	1-10	1
B	11-15	2
C	16-25	3
D	26-30	6
- 7) There are internal choices in Q. Nos. 28 and 30.
- 8) Write on both sides of the pages of your answer-book. If any rough work is to be done, do it on last pages of the answer-book and cross with slant lines and write 'Rough Work' on them.
- 9) Draw the graph of Q. No. 26 on the graph paper.

PART-A

- 1. Find the HCF of integers 375 and 675 by the prime factorisation method.**

Solution:

Given that,

HCF of 96 and 404

Now, Prime Factorisation Method

$$375 = 3 \times 5^3$$

$$675 = 3^3 \times 5^2$$

$$HCL = 3 \times 5^2 = 75$$

2. Find 11th term of the A.P. -17, -12, -7

Solution:

Given that,

A.P. -17, -12, -7

$$a_2 = -12$$

$$a_1 = -17$$

$$\begin{aligned} d &= a_2 - a_1 \\ &= -12 - (-17) \\ &= 5 \end{aligned}$$

$$\begin{aligned} a_{11} &= a + 10d \\ &= -17 + 10(5) \\ &= -17 + 50 \\ &= 33 \end{aligned}$$

Hence, the 11th terms of A.P = 33.

3. If $\cos A = \frac{12}{13}$, then calculate $\cot A$.

Solution:

Given that,

$$\cos A = \frac{12}{13} = \frac{B}{H}$$

$$B = 12, H = 13$$

$$P^2 = H^2 - B^2$$

$$= 13^2 - 12^2$$

$$= 169 - 144$$

$$P = \sqrt{25}$$

$$P = 5$$

So,

$$\cot A = \frac{12}{5}$$

4. Express the trigonometric ratio $\tan A$ in terms of $\sec A$.

Solution:

Consider

$$\tan^2 A = \sec^2 A - 1$$

$$\tan A = \sqrt{\sec^2 A - 1}$$

5. The area of two similar triangles are in ratio 16:81. Find the ratio of its sides.

Solution:

Given that,

$$\triangle ABC \sim \triangle PQR$$

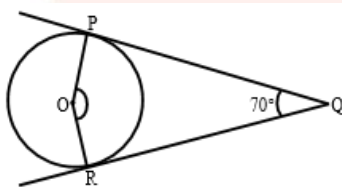
$$\frac{ar(ABC)}{ar(PQR)} = \left(\frac{AB}{PQ}\right)^2 \quad (\text{Theorem 6.6})$$

$$\frac{16}{81} = \left(\frac{AB}{PQ}\right)^2$$

$$\frac{AB}{PQ} = \sqrt{\frac{16}{81}} = \frac{4}{9}$$

$$AB : PQ \Rightarrow 4 : 9$$

6. In the given figure, O is the center of a circle and two tangents QP and QR are drawn on the circle from a point Q lying outside the circle. Find the value of angle POR.



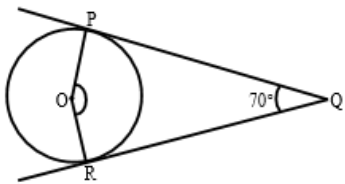
Solution:

Consider,

Angle between Radius and tangent

$$\angle P = 90^\circ$$

$$\angle R = 90^\circ$$



from $\square OPQR$,

$$\angle POR + \angle PQR = 180$$

$$\angle POR + 70 = 180$$

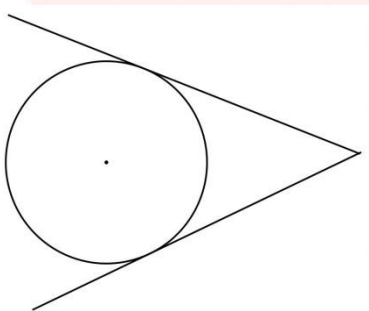
$$\angle POR = 180 - 70$$

$$\angle POR = 110$$

- 7. How many tangents can be drawn on the circle of radius 5 cm from a point lying outside the circle at distance 9 cm from the center.**

Solution:

Consider



Hence, the two tangent drawn the circle at distance 9 cm from the center.

8. Find the radius of that circle whose area is 616 cm^2 .

Solution:

Area of circle = 616

$$\pi r^2 = 616$$

$$\frac{22}{7} \times r^2 = 616$$

$$r^2 = 616 \times \frac{7}{22}$$

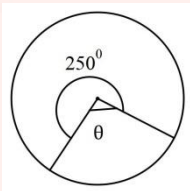
$$r^2 = 196$$

$$r = \sqrt{196}$$

$$= 14$$

9. If the angle of major sector of a circle is 250° . Then find the angle of minor sector.

Solution:



Angle of minor sector = $360 - 250 = 110$

10. A coin is tossed once. Find the probability that it is not a tail.

Solution:

Given that,

Total number of event = 2

$$P(E) = \frac{\text{no. of favorable event}}{\text{total event}} = \frac{1}{2}$$

PART-B

11. If the middle point of two points A(-2, 5) and B(-5, y) is $\left(\frac{-7}{2}, 3\right)$ then find the distance between points A and B.

Solution:

Given that,

$$A = (-2, 5), B = (-5, y)$$

And

$$\text{Mid point} = \left(\frac{-7}{2}, 3\right)$$

Mid point of coordinate

$$y = \left(\frac{y_2 + y_1}{2}\right)$$

$$y = \frac{y_2 + y_1}{2}$$

$$3 = \frac{y + 5}{2}$$

$$y + 5 = 6$$

$$y = 6 - 5$$

$$y = 1$$

Hence, the required value of $y = 1$

12. The total surface area of a Solid hemisphere is 462 cm^2 . Find its radius.

Solution:

Let r = radius of Solid hemisphere

Total surface area of hemisphere = 462 cm^2

$$3\pi r^2 = 462$$

$$3 \times \frac{22}{7} \times r^2 = 462$$

$$r^2 = \frac{462 \times 7}{3 \times 22}$$

$$r^2 = 49$$

$$r = 7 \text{ cm}$$

Hence, the radius of Solid hemisphere $r = 7 \text{ cm}$.

13. Per day expenses of 25 families of the frequency distribution of a Dhani of a village is given as follows.

Pre day expense (In Rs.)	25-35	35-45	45-55	55-65	65-75
Number of families	3	7	6	6	3

Find the mean expense of families by Direct Method.

Solution:

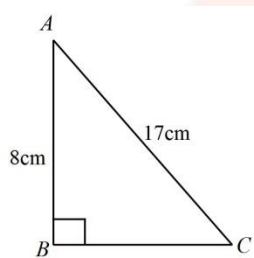
Pre day expense (In Rs.)	Frequency f_i	Mid value x_i	$(f_i \times x_i)$
25-35	3	30	90
35-45	7	40	280
45-55	6	50	300
55-65	6	60	360
65-75	3	70	210
	$\sum f_i = 30$		$\sum (f_i \times x_i) = 1240$

Thus,

$$\begin{aligned} \text{mean}(\bar{x}) &= \frac{\sum (f_i \times u_i)}{\sum f_i} \\ &= \frac{1240}{25} = 49.6 \end{aligned}$$

14. For traffic control, a CCTV camera is fixed on a 8m straight pole. The camera can see 17m distance sight line from the top. Find the area visible by the camera around the pole?

Solution:



$$\begin{aligned} B^2 &= H^2 - P^2 \\ &= 17^2 - 8^2 \\ &= 289 - 64 \\ &= 225 \\ b &= \sqrt{225} \\ &= 15 \end{aligned}$$

$$\text{Area of triangle ABC} = \frac{1}{2} \times 15 \times 8 = 60$$

15. A Motor car travels 175 km distance from a place A to place B, at a uniform speed 70km/hr passes through all ten green traffic signals. Due to heavy traffic it stops for one minute at first signal, 3 minutes at second signal, 5 minutes at third signal and so on stops for 19 minutes at tenth signal. How much total time it takes to reach at the place B. Solve by suitable mathematical method.

Solution.

Simplify the expression,

We know that

$$\text{Time} = \frac{\text{distance}}{\text{speed}}$$

Then,

$$\text{speed}(s) = 65 \text{ km/h}$$

$$\text{distance}(d) = 260 \text{ km}$$

$$t = \frac{175}{70}$$

$$t = 2 \text{ h } 30 \text{ min}$$

$$S_n = \frac{n}{2}[a + l]$$

$$= \frac{10}{2}[1 + 19]$$

$$= 5 \times 20$$

$$= 100 \text{ min}$$

$$\therefore 60 \text{ min} + 40 \text{ min}$$

$$= 1 \text{ h } 40 \text{ min}$$

Total time take

$$= 2 \text{ h } 30 \text{ min} + 1 \text{ h } 40 \text{ min}$$

$$= 4 \text{ h } 10 \text{ min}$$

PART-C

16. Prove that $\sqrt{6}$ is an irrational number.

Solution:

Let $\sqrt{6}$ is a Rational no.

$$\sqrt{6} = \frac{a}{b} \text{ then, } b \neq 0$$

Now, squaring both sides

$$(\sqrt{6})^2 = \left(\frac{a}{b}\right)^2$$

$$\frac{6}{1} = \frac{a^2}{b^2}$$

$$6b^2 = a^2 \text{ -----(i)}$$

$\therefore a^2$, div by 6

$\therefore a$ also divide by 6 then,

$$\frac{a}{6} = \frac{c}{1}$$

$$a = 6c$$

Put the value of a in equation (i)

$$6b^2 = (6c)^2$$

$$6b^2 = 36c^2$$

$$b^2 = 6c^2$$

$$c^2 = \frac{b^2}{6}$$

$\therefore b^2$ divisional by 6

$\therefore b$ also divisional by 6

So, $\sqrt{6}$ is a irrational.

17. Find the zeroes of the quadratic polynomial $x^2 + x - 2$, and verify the relationship between the zeroes and coefficients.

Solution:

Consider

$$x^2 + x - 2$$

$$x^2 + 2x - x - 2$$

$$x(x+2) - 1(x+2)$$

$$(x+2)(x-1)$$

$$\therefore x+2 = 0$$

$$x = -2$$

and,

$$x-1 = 0$$

$$x = 1$$

Now, verify the value

$$\alpha + \beta = \frac{-b}{a}$$

$$-2 + 1 = \frac{-1}{1}$$

$$-1 = -1$$

and,

$$\alpha \times \beta = \frac{c}{a}$$

$$(-2) \times 1 = \frac{-2}{1}$$

$$-2 = -2$$

18. Find the sum of first 15 terms of an A.P. whose 5th and 9th terms are 26 and 42 respectively.

Solution:

$$5^{\text{th}} \text{ term} = 26$$

So,

$$a_5 = 26$$

Then,

$$a + 4d = 26 \text{ --- (i)}$$

Similarly,

$$a + 8d = 42 \text{ --- (ii)}$$

Subtracting 2 from 1 we get,

$$4d = 16$$

$$d = 4$$

Substituting value of d in (i)

$$a + 4d = 26$$

$$a + 4(4) = 26$$

$$a + 16 = 26$$

$$a = 26 - 16$$

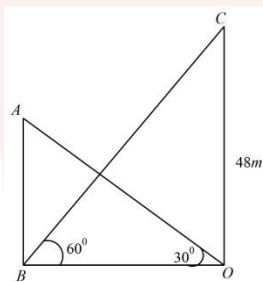
$$a = 10$$

Now,

$$\begin{aligned} S_{15} &= \frac{n}{2}[2a + (n-1)d] \\ &= \frac{15}{2}[2(10) + (15-1)d] \\ &= \frac{15}{2}[20 + 14(4)] \\ &= \frac{15}{2}[20 + 56] \\ &= \frac{15}{2} \times 76 \\ &= 15 \times 38 \\ &= 570 \end{aligned}$$

19. The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 48 meters high, find the height of the building.

Solution:



Let AB be the building and CD be the tower

In $\triangle CDB$,

$$\frac{CD}{BD} = \tan 60^\circ$$

$$\frac{48}{BD} = \sqrt{3}$$

$$BD = \frac{48}{\sqrt{3}}$$

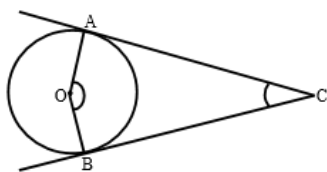
In $\triangle ABD$

$$\frac{AB}{BD} = \tan 30^\circ$$

$$AB = \frac{48}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{48}{3} = 16$$

Hence, the height of the building is 16m.

20. In the given figure, O is the center of a circle and two tangents CA, CB are drawn on the circle from a point C lying outside the circle. Prove that $\angle AOB$ and $\angle ACB$ are supplementary



Solution:

Consider the $\triangle OCA$ & $\triangle OBC$

$CA = CB$ (Tangents drawn from an external point are equal)

$OA = OB$ (Radii of the same circle)

Therefore, $\triangle OCA \cong \triangle OBC$ (SSS congruence criterion)

Hence,

$$\angle OCA = \angle OCB$$

$$\angle AOC = \angle BOC$$

Also,

$$\angle ACB = 2\angle OCA \text{ ----- (i)}$$

$$\angle AOB = 2\angle AOC \text{ ----- (ii)}$$

In the Right angled Triangle $\triangle OAC$

$$\angle AOC + \angle OCA = 90^\circ$$

$$\angle AOC = 90^\circ - \angle OCA \text{ ----- (iii)}$$

Multiplying the equation (3) by 2,

$$2\angle AOC = 180^\circ - 2\angle OCA$$

By Substituting (1) and (2) in the equation above,

$$\angle AOB = 180^\circ - \angle ACB$$

$$\angle AOB + \angle ACB = 180^\circ$$

21. Construct a triangle with sides 4cm, 5cm and 7cm and then another triangle whose sides are $\frac{3}{4}$ of the corresponding sides of the first triangle.

Solution:

Step-1 First of all we draw a line segment

$$AB = 4cm$$

Step-2 With A as centre and radius

$$AC = 5cm$$

Step-3 With B as centre and radius

$$BC = 7cm$$

Draw an arc, intersecting the arc drawn in step-2 at C.

Step-4 Join AC and BC to obtain

$$\triangle ABC$$

Step-5 Below AB, makes an acute angle

$$\angle BAX = 60^\circ$$

Step-6 AX, mark off four points

$$A_1, A_2, A_3 \text{ and } A_4$$

Such that,

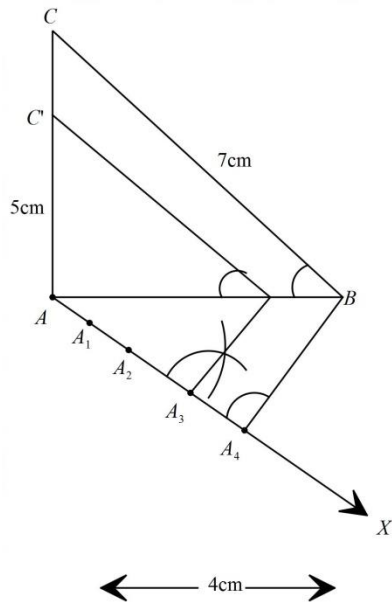
$$AA_1 = A_1A_2 = A_2A_3 = A_3A_4$$

Step-7 join BA_4 and draw a line through A_3 parallel to BA_4 to intersect AB at point B'.

Step -8

Draw a line through B' parallel to the line BC to intersect AC at C'.

$\triangle A'B'C'$ is the required triangle.



Then, the construction can prove that

$$AB' = \frac{3}{4} AB, B'C' = \frac{3}{4} BC, AC' = \frac{3}{4} AC$$

By construction, we have $B'C' \parallel BC$

$$\therefore \angle A$$

$$B'C'$$

$$= \angle ABC \text{ (Corresponding angles)}$$

In $\triangle AB'C'$ and $\triangle ABC$,

$$\angle AB'C'$$

$$= \angle ABC \text{ (proved above)}$$

$$\angle BA'C'$$

$\sim \triangle ABC$ (AA similarity criterion)

$$\Rightarrow \frac{AB'}{AB} = \frac{B'C'}{BC} = \frac{AC'}{AC} \text{ ----- (i)}$$

In $\triangle AA_3B'$ and $\triangle AA_4B$

$\angle A_3AB' = \angle A_4AB$ (common factor)

$\angle AA_3B' = \angle AA_4B$ (corresponding angles)

$\therefore \triangle AA_3B' \sim \triangle AA_4B$ (AA similarity criterion)

$$\Rightarrow \frac{AB'}{AB} = \frac{AA_3}{AA_4} \text{----- (ii)}$$

From equation (i) and (ii), we have

$$\frac{AB}{AB} = \frac{BC}{BC} = \frac{AC'}{AC} = \frac{3}{4}$$

$$\Rightarrow AB' = \frac{3}{4}AB, B'C' = \frac{3}{4}BC, AC' = \frac{3}{4}AC$$

22. If an arc of a circle subtends an angle of 45° at the centre and if the area of minor sector is 77cm^2 , then find the radius of the circle.

Solution:

Given that,

Area of minor sector = 77cm^2

Circle subtends an angle of $= 45^\circ$

Let, circle of radius = r

Then,

$$\frac{\pi r^2 \theta}{360} = 77$$

$$\Rightarrow \frac{22}{7} \times r^2 \times \frac{45}{360} = 77$$

$$\Rightarrow r^2 = 196$$

$$r = 14\text{cm}$$

Hence, the radius of circle = 14 cm.

23. Seven spheres of equal radii are made by melting a silver-cuboid of dimensions 8cm×9cm×11cm- Find the radius of a silver sphere.

Solution:

Consider

Silver-cuboid of volume = volume of spheres are Equal to 7 radius

Let,

1 sphere of radius r

$$\therefore 8 \times 9 \times 11 = 7 \times \frac{4}{3} \pi r^3$$

$$\Rightarrow r^3 = \frac{8 \times 9 \times 11 \times 3}{7 \times 4} \times \frac{7}{22}$$

$$\Rightarrow r^3 = 27$$

$$\therefore r = 3$$

Hence, the radius of a silver sphere = 3.

24. The following table shows the marks obtained by 50 students in mathematics of class X in a school.

Obtained Marks	20-30	30-40	40-50	50-60	60-70	70-80
Number of students	5	9	8	12	13	3

Solution:

Obtained Marks	Number of students f_i	cf
20-30	5	5
30-40	9	14
40-50	8	22
50-60	12	34
60-70	13	47
70-80	3	50
	$\sum f_i = 50$	

$$m = \frac{N}{2} = \frac{50}{2} = 25\text{th term}$$

Then,

$$\begin{aligned} M &= l_1 + \frac{\frac{N}{2} - cf}{f} \times h \\ &= 50 + \frac{25 - 22}{12} \times 10 \\ &= 50 + \frac{3}{12} \times 10 \\ &= 50 + \frac{5}{2} \\ &= \frac{55}{2} = 52.5 \end{aligned}$$

- 25. A piggy bank contains hundred coins of Rs. 1, twenty five coins of Rs. 2, fifteen coins of Rs. 5 and ten coins of Rs. 10. If it is equally likely that one coin will fall, when the bank is turned upside down, what is the probability that the coin –**
(i) Will be a Rs. 2 coin? (ii) Will not be a Rs. 5 coin?

Solution:

Given that,

No. of 1 Rs coin = 100

No. of 2 Rs coin = 25

No. of 5 Rs coin = 15

No. of 10 Rs coin = 10

Total number of coin = 150

(i)

$$P(E) = \frac{\text{fav. event}}{\text{total event}}$$

$$P(E) = \frac{25}{150} = \frac{1}{6}$$

(ii)

$$P(E) = \frac{\text{fav. event}}{\text{total event}}$$

$$P(\text{will be a Rs 5 coin}) = \frac{15}{150} = \frac{1}{10}$$

$$P(E) + P(E \text{ not}) = 1$$

$$P(E \text{ not}) = 1 - \frac{1}{10}$$

$$= \frac{10-1}{10}$$

$$= \frac{9}{10}$$

PART-D

26. The cost of 2 exercise books and 3 pencils is Rs.17 and the cost of 3 exercise books and 4 pencils is Rs. 24. Formulate the problem algebraically and solve it graphically.

Solution:

Let cost of exercise books be x Rs.

And, let cost of Pencil be y Rs.

So,

$$2x + 3y = 17 \text{ ----- (i)}$$

$$3x + 4y = 24 \text{ ----- (ii)}$$

Multiplying 3 in equation (i) and multiplying 2 in equation (ii). We obtain

$$6x + 9y = 51 \text{ ----- (iii)}$$

$$6x + 8y = 48 \text{ ----- (iv)}$$

Subtracting equation (iii) and (iv)

$$6x + 9y = 51$$

$$6x + 8y = 48$$

$$\begin{array}{r} - \quad - \quad - \\ \hline \end{array}$$

$$y = 3$$

Putting the value y in equation (i)

$$2x + 3y = 17$$

$$2x + 3(3) = 17$$

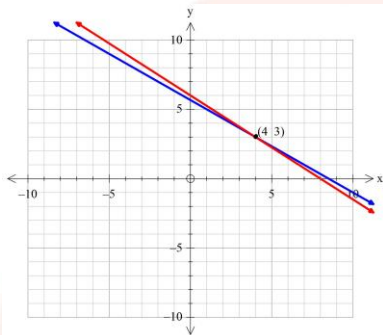
$$2x + 9 = 17$$

$$2x = 17 - 9$$

$$2x = 8$$

$$x = \frac{8}{2}$$

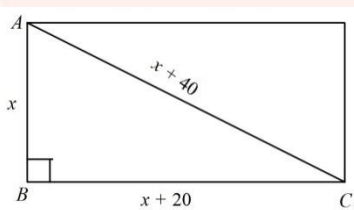
$$x = 4$$



27. The diagonal of a rectangular field is 40 meters more than the shorter side. If longer side is 20 meters more than the shorter side, find the sides of the field.

Solution:

Let,



Smallest side = x

Longer side = $x + 20$

And diagonal of a rectangular = $x + 40$

Form $\triangle ABC$

$\angle B = 90^\circ$ (Rectangle)

$$H^2 = P^2 + B^2$$

$$(x + 40)^2 = x^2 + (x + 20)^2$$

$$x^2 + 2(x)(40) + (40)^2 = x^2 + x^2 + 2(x)(20) + (20)^2$$

$$80x + 1600 = x^2 + 40x + 400$$

$$\Rightarrow x^2 + 40x + 400 - 80x - 1600 = 0$$

$$x^2 - 40x - 1200 = 0$$

$$x^2 - 60x + 20x - 1200 = 0$$

$$x(x - 60) + 20(x - 60) = 0$$

$$(x + 20) + (x - 60) = 0$$

then,

$$x + 20 = 0$$

$$x = -20$$

so,

$$x - 60 = 0$$

$$x = 60$$

Hence, the longer side = $60 + 20 = 80$

OR,

A Pole has to be erected at a point on the boundary of a circular park of diameter 17 meters in such a way that the differences of its distances from two diametrically opposite fixed gates A and B on the boundary is 7 meters. Is it possible to do so? If yes, at what distances from the two gates should the pole be erected?

Solution:

Let,

Circular park of diameter = 17 m

Form $\triangle ABC$

$\angle C = 90^\circ$ (Angle in semi circle)

$$H^2 = P^2 + B^2$$

$$17^2 = x^2 + (x+7)^2$$

$$289 = x^2 + x^2 + 2(x)(7) + (7)^2$$

$$2x^2 + 14x + 49 - 289 = 0$$

$$2x^2 + 14x + 49 - 289 = 0$$

$$x^2 + 7x - 120 = 0$$

$$x^2 + 15x - 8x - 120 = 0$$

$$x(x+15) - 8(x+15) = 0$$

$$(x+15)(x-8) = 0$$

so,

$$x+15 = 0$$

$$x = -15$$

and,

$$x-8 = 0$$

$$x = 8$$

Distance from the first gate = 8m

And 2nd distance from the gate = $8+7 = 15$ m

28. (i) If $\cos 3A = \sin(A - 34^\circ)$, where A is an acute angle, find the value of A.

Solution:

We have,

$$\cos 3A = \sin(A - 34^\circ)$$

$$\Rightarrow \cos 3A = \cos(90 - (A - 34))$$

$$3A = 90 - A + 34$$

$$4A = 124$$

$$A = \frac{124}{4}$$

$$A = 31$$

(ii) Prove the following identity, where the angles involved are acute angles for which the expression is defined.

$$\frac{1 + \cot^2 A}{1 + \tan^2 A} = \left(\frac{1 - \cot A}{1 - \tan A} \right)^2$$

Solution:

Consider,

Taking LHS,

$$\begin{aligned} & \frac{1 + \cot^2 A}{1 + \tan^2 A} \\ &= \frac{\operatorname{cosec}^2 A}{\sec^2 A} \\ &= \frac{1}{\frac{\sin^2 A}{1}} \\ &= \frac{\cos^2 A}{\sin^2 A} \\ &= \cot^2 A \end{aligned}$$

Taking RHS

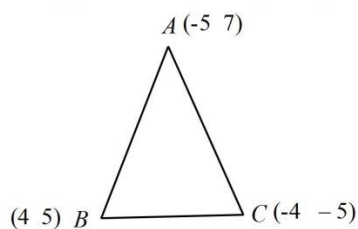
$$\begin{aligned} & \left(\frac{1 - \cot A}{1 - \tan A} \right)^2 \\ &= \left(\frac{1 - \cot A}{1 - \frac{1}{\cot A}} \right)^2 \\ &= \left(\frac{1 - \cot A}{\frac{(\cot A - 1)}{\cot A}} \right)^2 \\ &= \left(\frac{\cos A(1 - \cot A)}{(\cot A - 1)} \right)^2 \\ &= (-\cot A)^2 \\ &= \cot^2 A \end{aligned}$$

Hence, LHS = RHS.

29. Find the area of that triangle whose vertices are (-5, 7), (4, 5) and (-4, -5).

Solution:

Consider

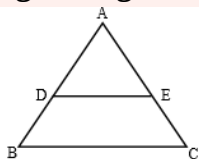


Area of triangle

$$\begin{aligned}
 &= \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \\
 &= \frac{1}{2} [(-5)(5 - (-5)) + 4((-5) - 7) + (-4)(7 - 5)] \\
 &= \frac{1}{2} [(-5) \times 10 + 4 \times (-12) + (-4) \times (7 - 5)] \\
 &= \frac{1}{2} [-50 - 48 - 8] \\
 &= \frac{1}{2} (-106) \\
 &= -53
 \end{aligned}$$

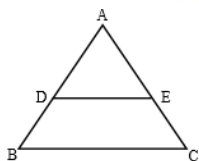
So, the area of triangle = 53 unit²

30. In the given figure ABC is a triangle. If $\frac{AC}{AE} = \frac{AB}{AD}$, then prove that $BC \parallel DE$



Solution:

Consider



Given that,

$$\frac{AC}{AE} = \frac{AB}{AD}$$

Construction – draw $DE \parallel BC$

Proof:

$$\frac{AC}{AE} - 1 = \frac{AB}{AD} - 1$$

$$\frac{AC - AE}{AE} = \frac{AB - DB}{AD}$$

$$\frac{CE}{AE} = \frac{DB}{AD}$$

$$\frac{AE}{EC} = \frac{AD}{DB}$$

$\therefore DE \parallel BC$ (By converse of BPT)

OR,

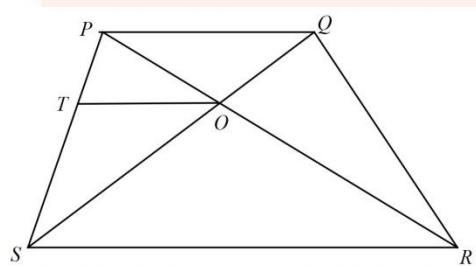
The diagonals of a quadrilateral PQRS intersect each other at the point O such that

$$\frac{PO}{OR} = \frac{OQ}{OS}$$

Show that PQRS is a trapezium.

Solution:

Consider



In $\triangle PQS$, $OE \parallel AB$

By using basic proportionality theorem, we get

$$\frac{AE}{ED} = \frac{BO}{OD} \text{-----(i)}$$

However, it is given that,

$$\frac{PO}{OR} = \frac{OQ}{OS} \text{-----(ii)}$$

From equation (i) and (ii) we obtain

$$\frac{PT}{TS} = \frac{PO}{OR}$$

$\Rightarrow OT \parallel RS$ [by the converse of basic proportionality theorem]

$\Rightarrow PQ \parallel OT \parallel RS$

$\Rightarrow PQ \parallel RS$

$\therefore PQRS$ is a trapezium.

